

Facts and Opinion

Fact is true, can be proved with evidence.

Opinion is a personal belief or view and may differ from person to person.

Partiality

Tendency to favour unfairly It leads to biased decisions.

Preference

- * Choosing one thing over another based on personal liking.
- * It is a personal choice and is not always unfair.

Prejudice (Bias)

Prejudice means preconceived opinion

A fixed opinion formed without proper knowledge or evidence.

It can result in discrimination and unfair treatment.

Common examples:

- * Racism
- * Sexism
- * Ageism
- * Classism
- * Xenophobia
- * Religious prejudice
- * Nationalism

Bias in Data Science

Bias is a systematic error in data or analysis that affects results.

It usually occurs due to:

- * Poor sampling
- * Incomplete data
- * Estimation errors
- * Biased data can produce unfair AI and Machine Learning models.

Identifying Bias

Common types of bias are:

- * Selection Bias
- * Linearity Bias
- * Confirmation Bias
- * Recall Bias
- * Survivor Bias
- * Availability Bias

Statistical bias:

Statistical bias is the systematic difference between an estimated value and the true value of a population parameter. It occurs when a method of collecting data, analysing data, or estimating a parameter consistently produces results that are too high or too low.

Selection Bias

- * Happens when data is selected in a non-random way.
- * Some groups are over-represented or under-represented.
- * It produces inaccurate and misleading results.
- * Proper sampling methods help reduce selection bias.

Linearity Bias

is the belief that two variables always have a straight-line relationship.

- * In reality, many relationships are *non-linear*.
- * Example: Fuel efficiency and engine size do not always change in a straight-line pattern.
- * *Graphs* help identify the actual relationship and reduce linearity bias.

Confirmation Bias

- * *Confirmation Bias* is the tendency to accept information that supports our beliefs and ignore evidence against them.
- * It can lead to wrong conclusions and poor decisions.
- * Example: Assuming ice cream sales cause crime because both increase in summer.
- * *How to reduce it:* Keep an open mind, examine opposing evidence, and question your assumptions.

Recall Bias

- * *Recall Bias* occurs when data is remembered or labelled inconsistently.
- * It often happens during data collection or data labeling.
- * It reduces the accuracy of the dataset.
- * Example: Similar damaged laptops being labelled differently.

Survivor Bias

- * *Survivor Bias* means focusing only on successful cases while ignoring failures.
- * This gives a misleading picture of reality.
- * Example: Studying only successful companies without considering failed ones.

Availability Bias

- * *Availability Bias* occurs when decisions are made using only easily available or recent information.
- * Important information may be ignored.
- * To avoid it, collect complete data and consider all relevant evidence before making decisions.